

1) Age of child and photograph Preference

Sol Given data

Age of child	A	B	C	Total
5-6 years	18	22	20	60
7-8 years	2	28	40	70
9-10 years	20	10	40	70
Total	40	60	100	200

H_0 : Age and photograph

H_1 : Age and photograph, Preference are Significantly related

R-Program:

```
data <- matrix(c(18, 22, 20
```

```
2, 28, 40
```

```
20, 10, 40)
```

```
nrow = 3, byrow = TRUE)
```

```
colnames(data) <- c("A", "B", "C")
```

```
rownames(data) <-
```

```
c("5-6", "7-8", "9-10")
```

```
data
```

Output

$$x\text{-squared} = 29.603$$

$$P\text{-value} = 5.895 - 0.6$$

$$dp = 4$$

Covariance b/w B and C

$$B = 22, 28, 10$$

$$C = 20, 40, 40$$

R-Code:

$$B \leftarrow C(22, 28, 10)$$

$$C \leftarrow C(20, 40, 40)$$

$$\text{cov}(B, C)$$

Output

$$(1) - 20$$

$$\text{Sample Covariance} = -20$$

2)

Given data

Player	Points	Player	Points
P ₁	12	P ₉	11
P ₂	15	P ₁₀	35
P ₃	14	P ₁₁	16
P ₄	10	P ₁₂	17
P ₅	18	P ₁₃	19
P ₆	20	P ₁₄	21
P ₇	22	P ₁₅	23
P ₈	13		

Box plot values

Age:

• minimum = 23

• $Q_1 = 39.5$

• median = 51

$Q_3 = 56.75$

• maximum = 61

% fat

minimum = 7.8

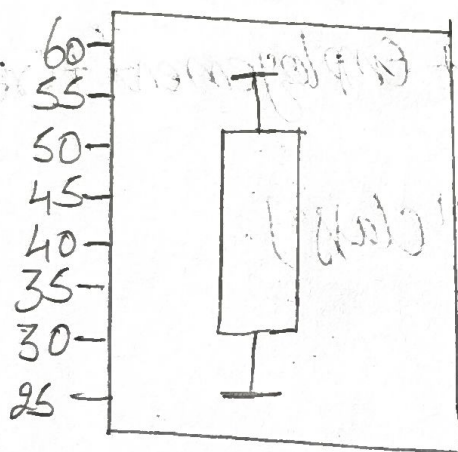
$Q_1 = 26.675$

median = 30.7

$Q_3 = 33.925$

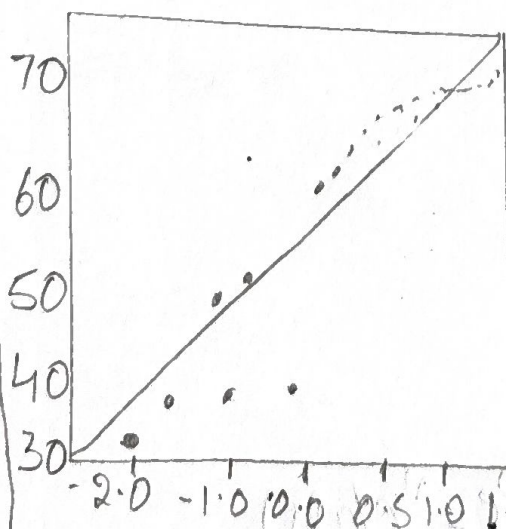
maximum = 42.5

Box Plot of Age:



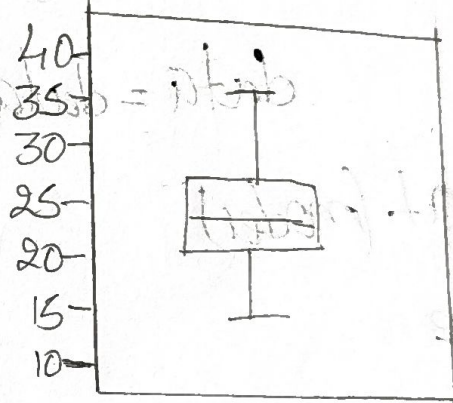
Age

a-a Plot of Age



Theoretical Quantities

Box Plot of Body fat %



$Q_1 = 13$

$Q_3 = 21$

$IQR = Q_3 - Q_1$

$IQR = 21 - 13 = 8$

Upper limit

$Q_3 + 1.5 \times IQR$

$= 21 + 1.5(8)$

$= 33$

3. R-Program

Install - Package ("x part")

Install . Package("xpart.plot")

library(xpart)

library(xpart.plot)

Build Decision tree

model <- xpart(loan ~ Age + Income + Employment + Credit,

data = data, method = "class")

Print(model)

Plot the tree

xpart.plot(model)

type = "n"

Extra = 100,

fallen.leaves = TRUE

Predict loan approval

Pred <- Predict(model, data, type = "class")

Confusion matrix

table(Actual = data\$loan

Predicted = Pred

4) Class A

76, 35, 47, 64, 95, 66, 89, 36, 84

Class B

51, 56, 84, 60, 59, 70, 63, 66, 50

i) mean, median & Range

Class A

$$\frac{592}{9} = 65.78 \rightarrow \text{mean}$$

median = 6

Range

$$95 - 35 = 60$$

Class B

$$\frac{559}{9} = 62.11$$

median = 60

Range

$$84 - 50 = 34$$

ii) Boxplot

Class A <-

c(76, 35, 47, 64, 95, 66, 39, 36, 84)

Class B <-

c(51, 56, 84, 60, 59, 70, 63, 66, 50)

boxplot(Class A, Class B,

names = c("Class A", "Class B")

main = "Box Plot of Maths Exam Scores",
ylab = "marks")

5) Given data

Age	Total
23	9.5
23	26.5
27	7.8
27	17.8
39	31.4
41	25.9
47	27.4
49	27.2
50	31.2
52	34.8
54	43.5
54	28.8
56	33.4
57	30.2
58	34.1
58	32.9
60	41.2
61	35.7

Age:

mean = 46.44

median = 51

Standard deviation = 13.27

% fat

• mean = 28.78

• median = 30.70

Standard deviation = 9.9